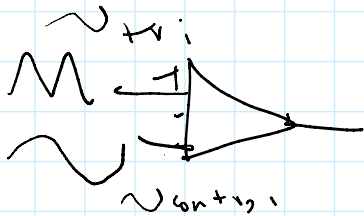


Recall DC/AC converters (inverters)

USL / CSL $\rightarrow 1\phi$
 $\rightarrow 3\phi$

USL $1\phi \rightarrow$ PWM $\rightarrow$ SPWM (sinusoidal PWM)



$v_{control,i} > v_{tr,i}$

S_A on

S'_A off

S_B off

S'_B on

S_B on (S'_B off)

S'_B on (S_B off)

$v_{control,i} > v_{tr,i}$

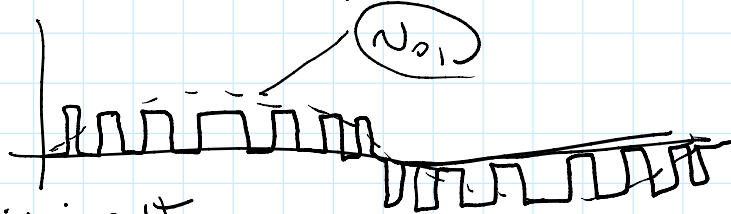
Example: In a USL- 1ϕ -SPWM, prove that

$$v_o = m_a v_i \sin \omega t$$

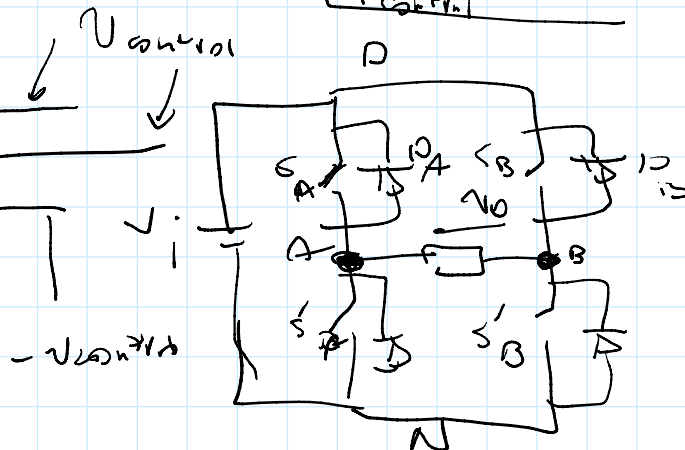
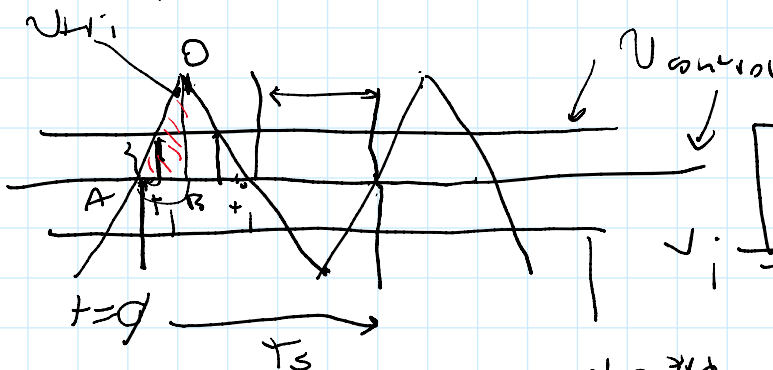
m_a = modulation index

$$\frac{\hat{v}_{control}}{\hat{v}_{tr,i}}$$

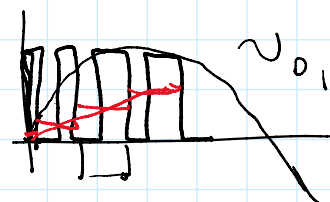
v_i : input



assumption: $m_a < 1$ (linear), $m_f = \frac{f_{tr,i}}{f_{control}}$ high



$$v_o = v_{AN} - v_{BN}$$



$$S_A: v_{control,i} > v_{tr,i}$$

$$t_{on} = 2t_1 + \frac{T_s}{2}$$

$$\Delta_{OAB}: \frac{t_1}{T_s/4} = \frac{v_{control,i}}{\hat{v}_{tr,i}}$$

$$t_1 = \frac{v_{control,i}}{\hat{v}_{tr,i}} \times \frac{T_s}{4}$$

$$t_i = \frac{V_{control}}{V_{Ti}} \times \frac{T_s}{4}$$

$$t_{on} = \frac{V_{control}}{\hat{V}_{Ti}} \times \frac{T_s}{2} + \frac{T_s}{2} = \frac{T_s}{2} \left(1 + \frac{V_{control}}{\hat{V}_{Ti}} \right), D = \frac{t_{on}}{T_s}$$

$$D_{on} = \frac{1}{2} \left(1 + \frac{V_{control}}{\hat{V}_{Ti}} \right) \quad D_{on_{SB}} = 1 - D_{on_{SA}}$$

$$M_a < 1$$

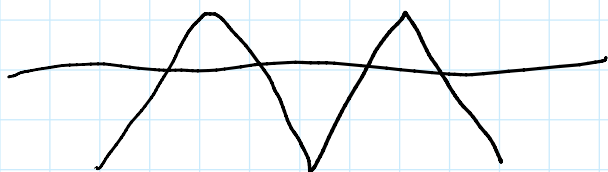
$$V_o = (D_1 - D_2) V_i = (2D_{on_{SA}} - 1) V_i = \frac{V_{control} V_i}{\hat{V}_{Ti}}$$

$\downarrow$ local average $\downarrow$ $D_{on_{SA}}$

$$\Rightarrow V_o = M_a \hat{V}_{control} \sin \omega t$$

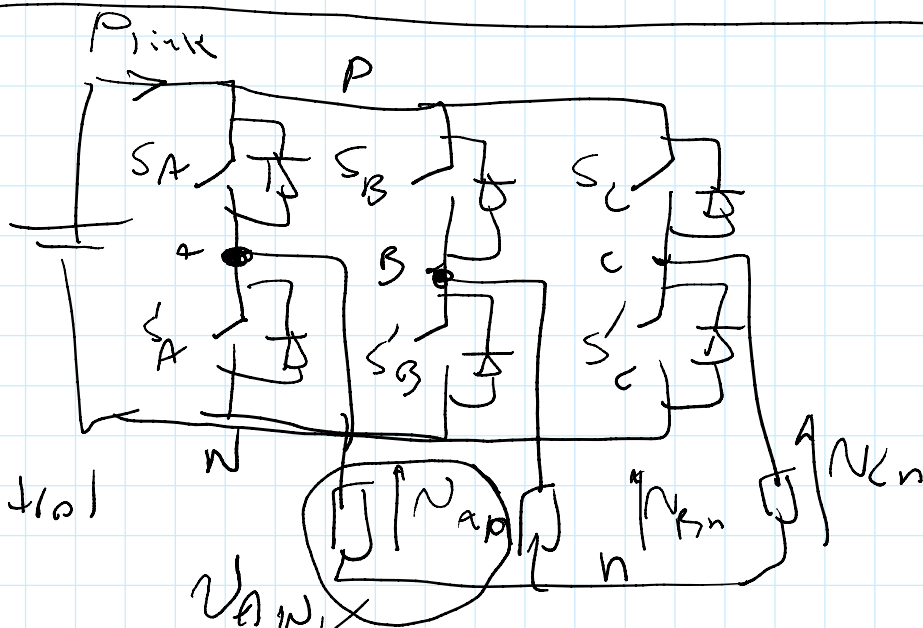
$$V_{control} = \hat{V}_{control} \sin \omega t$$

$\downarrow \hat{V}_{control}$

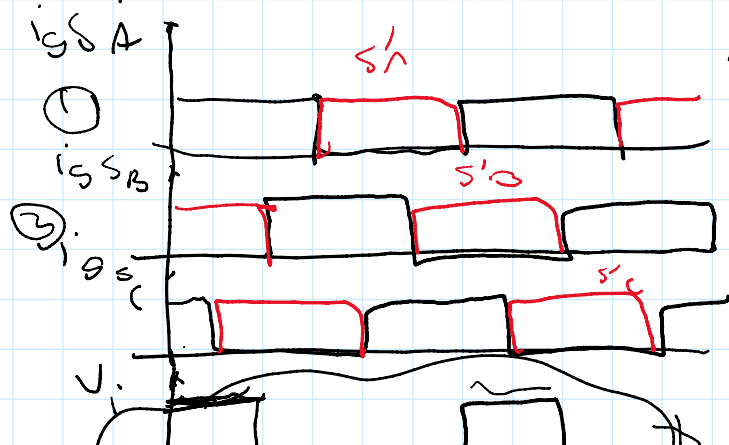


US-3Ø

V_i



square-wave control



$$V_{AN} = V_{AN} - V_{BN}$$

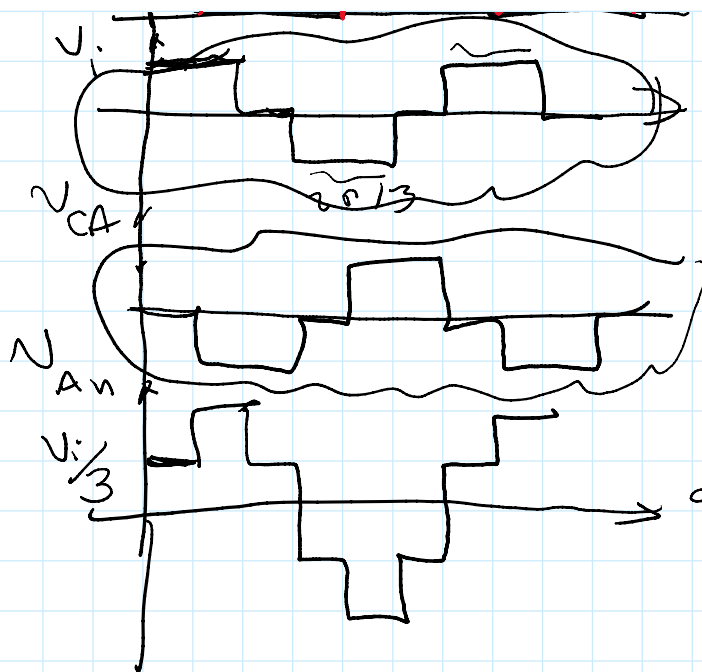
$$V_{BN} = V_{BN} - V_{CN}$$

$$V_{CN} = V_{CN} - V_{AN}$$

$$V_{AB} = V_{AN} - V_{BN}$$

$$V_{AN} = ?$$





V_{AB} 'V_{An}?' ←

$$\begin{aligned} V_{AB} &= V_{An} - V_{Bn} \\ V_{BC} &= V_{Bn} - V_{Cn} \\ V_{CA} &= V_{Cn} - V_{An} - V_{An} \\ V_{AB} - V_{CA} &= 2V_{An} - (V_{Bn} + V_{Cn}) \end{aligned}$$

assumption: $V_{An} + V_{Bn} + V_{Cn} = 0$

$$V_{AB} - V_{CA} = 3V_{An}$$

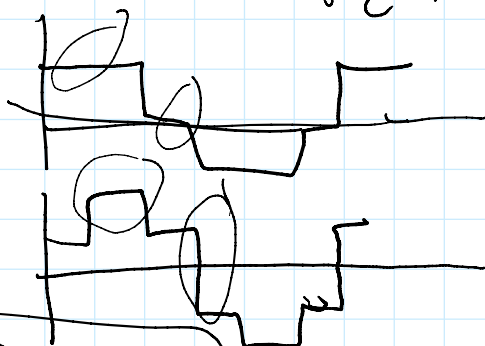
$$V_{An} = \frac{1}{3}(V_{AB} - V_{CA})$$

$$V_{eff} = V_{AB} = V_{BC} = V_{CA} = \left[\frac{1}{\pi} \int_0^{2\pi/3} V_i^2 d\omega t \right]^{1/2} = \sqrt{\frac{2}{3}} V_i \approx 0.8V_i$$

$$V_{AB} = \sum \frac{4V_i}{n\pi} \cos\left(\frac{n}{6}\right) \sin\left(n\omega t + \frac{n}{6}\right) = \sum \frac{4V_i}{n\pi} \frac{\sqrt{3}}{2} \sin\left(n\omega t + \frac{n}{6}\right)$$

$$V_{AB1} = \frac{2\sqrt{3}}{\pi} V_i \quad V_{AB} = \frac{\sqrt{3}}{\sqrt{2}} \frac{2}{\pi} V_i \quad \text{THD} = \frac{V_{AB1}}{V_{AB}} = 31\%$$

question



V_{AB}

V_{An}

THD

THD

$$V_{AB} = V_{An} - V_{Bn}$$

$$\text{or } V_{An} = \frac{V_{AB} + V_{CA}}{3}$$

$$\text{THD}_{AB} \leq \text{THD}_{An}$$

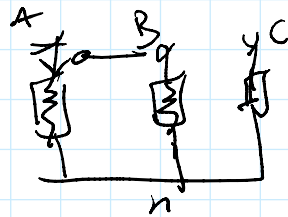
Example: $V_s = 3\phi$, $R = 10\Omega$ Y-connected, $f_s = 60\text{Hz}$

common PE: $V_s \angle 30^\circ$, $R = 10 \Omega$ Y-connected, $P_i = 60 \text{ W}$
 $V_i = 220 \text{ V}$

a) $V_{AB} = ?$ $V_{AB} = \sum \frac{4V_i}{n\pi} \cos \frac{n\pi}{6} \sin(n\omega t + \frac{n\pi}{6})$

$V_{AB} = 242.58 \sin(\omega t + \frac{\pi}{6}) - 48.52 \sin(5\omega t + \frac{5\pi}{6})$
 $- 321.66 \sin(7\omega t + \frac{7\pi}{6}) + \dots$

b) $i_{An} = ?$



$i_{An} = \frac{V_{An}}{R} = \frac{V_{An}}{10}$

$i_{An} = \left(\frac{V_{AB}(1)}{\sqrt{3}} \right) \angle -30^\circ \cdot \frac{1}{10}$

$i_{An} = 14 \sin(\omega t) - 2.8 \sin(5\omega t) + 2 \sin(7\omega t) + \dots$

c) $V_{\text{eff}} = ?$ $V_{\text{eff}} = 0.81 V_i = 179.83 \text{ V}$

d) $V_{\text{eff}} = ?$ $\frac{179.83}{\sqrt{3}} = 103.7 \text{ V}$

e) $\text{THD} = 31.1 \%$

f) $P_o = ?$ $P_o = 3 V_{\text{eff}} I = 3 \cdot 103.7 \times 10.37 = 3226.4 \text{ W}$

g) $I_{\text{C(AN)}} = ?$ $I_{\text{link}} = \frac{P_{\text{link}}}{V_i} = \frac{P_o}{V_i} = \frac{3226.4}{220} = 14.67 \text{ A}$
 IGBT

$I_{\text{C(AN)}} = \frac{14.67}{3} = 4.89$

x Safety Factor